This project uses AI to support laboratory personnel by interpreting OOC and trend-violation events after they have already been detected by rule-based logic in Power BI. AI is not used for compliance decisions or instrument control. Its role is strictly advisory and designed to help ensure that potential issues are not overlooked when staff are busy.

AI begins its work only after Power BI flags an OOC or trend violation. Within this boundary, AI is allowed to:

- Summarize the detected OOC or trend violation
- Compare the current event to the past 10-day baseline
- Incorporate environmental conditions such as temperature, humidity, and baseline stability
- Identify possible causes (e.g., instrument bias, standard cylinder degradation
- Suggest recommended next steps
- Recommend monitoring when the issue appears minor
- Recommend investigation when the issue appears significant
- Recommend notifying the lab manager
- Recommend temporarily pausing instrument use as a precaution

AI's purpose is to interpret, contextualize, and communicate the situation in a structured format.

AI is not allowed to:

- Declare an instrument officially failed
- Approve corrective actions
- Close investigations
- Override rule-based logic
- Make compliance or production decisions
- Decide to discard data
- Make any final determination about instrument status

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- AI can recommend, but it cannot decide.
- All final decisions remain with qualified staff.
- The lab manager has full and final authority over all decisions, including instrument status, investigations, and corrective actions.

- The lab tech must visually review the LTS graphs to confirm drift, bias, or standard degradation before any action is taken.

KPIs

This workflow is designed to improve how quickly and consistently LTS data issues are detected, reviewed, and acted on. The KPIs below measure the impact of the automated Power BI → Power Automate → AI escalation process compared to traditional manual review.

- Time-to-Detection (TTD)- how long it takes for an OOC or trend violation to be identified.
  - Baseline: Manual review can delay detection.
  - With workflow: Power BI flags issues immediately after refresh
- Time-to-Notification (TTN) - how long it takes for personnel to be notified after detection.
  - Baseline: Manual emails or missed reviews can delay notification.
  - With workflow: Power Automate sends notifications automatically.